Muhammad Rayyan Ahmed

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Cover Letter: Working Student – Machine Learning: Time-Series Analysis and Anomaly Detection (Vacancy-ID 6404)

DLR Institute of Data Science Jena 2026-11-03

Dear Prof. Thiel,

I am applying for the working student position in your Machine Learning group on time-series analysis and anomaly detection for glider flight data. Turning messy measurements into a clean, validated dataset and then testing whether a method really works is the kind of work I enjoy most.

I am an M.Sc. Data Science student at TU Ilmenau with a B.S. in Artificial Intelligence. I live and study in Ilmenau, so Jena is an easy commute, and the planned weekly hours fit well with my studies.

- **Evaluation,** for my RAG project DocuMind I designed an evaluation harness with hit rate, MRR and an LLM judge, and used it to trace both wrong answers to their cause
- **Data quality,** in industry I used SQL analysis, validation and reconciliation to check completeness and consistency of SAP data before reporting
- **Python pipelines,** I built preprocessing pipelines in Python and PyTorch, including a classifier that reached 94.11% accuracy on a 968-image test set, and DocuMind with 37 unit tests

I would start with a structured literature review, then explore the flight and sensor data, build cleaning and synchronisation workflows in NumPy and Pandas, and prototype and validate methods step by step with clear documentation. Time-series anomaly detection is newer to me than computer vision, and I am keen to learn it in a research setting.

I can start immediately. I would be glad to discuss the topic with you.

I work in English (IELTS 8.0) and am currently learning German (A1).